

Three Approaches for Building Information Modeling Library Transplant

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ABSTRACT

Building information modeling (BIM) library transplant is a framework proposed by the building informatics group at Yonsei University as a way to develop the details of a BIM model by automatically finding and replacing a low-level BIM library object with a matching high-level BIM library object in a donor BIM model. This paper introduces three BIM library transplant approaches: namely, attribute matching, semantic elaboration, and natural language-based architectural detailing through interaction with artificial intelligence (NADIA).

KEYWORDS : building information modeling (BIM), library transplant, graph neural network (GNN), semantic enrichment, semantic annotation, semantic elaboration, natural language-based architectural detailing through interaction with artificial intelligence (NADIA)

INTRODUCTION

Traditional automated design detail generation methods are mostly rule-based. Before non-programming rule-implementation methods, such as parametric modeling (Lee et al. 2006; Roller 1991; Shah 2001) and visual coding (Autodesk 2020; Grasshopper 2020) become prevalent, expert knowledge had to be elicited and hard-coded into a system. Although parametric modeling and visual coding lighten the burden to hard-code knowledge into a system (Lee 2022), traditional hard-coding or parametric modeling approaches are still limited in that they generally take a long time to elicit and translate expert knowledge into parametric relationships, thus, are inflexible in coping with users' unique preferences or new requirements.

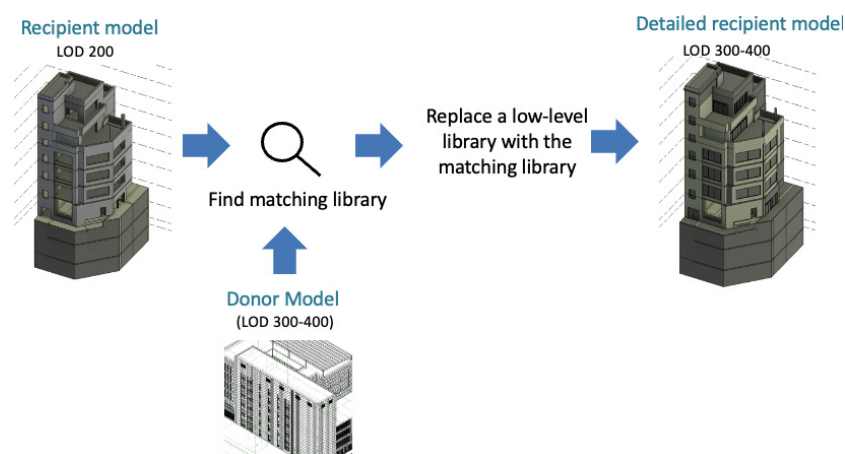


Figure 1. Overview of BIM library transplant

We propose “building information modeling (BIM) library transplant”—a new framework to automatically detail BIM models without generating knowledge-based rules (Figure 1). BIM library transplant develops details of a low-level BIM model (i.e., recipient model) by automatically finding and replacing a low-level BIM library object with a matching high-

level BIM library object in an existing BIM model (i.e., donor model). Similar to an organ transplant, the core of the BIM library transplant lies in a library matching algorithm. Depending on the matching algorithm, various BIM library transplant methods can be developed.

This paper introduces three approaches developed at the building informatics group at Yonsei University in Seoul, South Korea: namely, attribute matching, semantic elaboration, and Natural language-based Architectural Detailing through Interaction with Artificial intelligence (NADIA).

THREE APPROACHES FOR BIM LIBRARY TRANSPLANT

The first approach is *attribute matching* or *parameter matching*. Attribute matching compares the attributes of BIM object library objects of a low-level BIM model and those of a high-level BIM model, replacing low-level BIM objects with matching high-level BIM objects. A simple implementation example, illustrated in Figure 1, compares the object type (e.g., wall, window, slab, etc.) first and then considers the dimensional specifications and other attributes. For attribute matching, both rule-based and machine-learning approaches can be used to compare the similarity of BIM libraries.

The second approach, *semantic elaboration*, is similar to attribute matching but different in that it also compares the semantic relationships between an object of interest and its neighboring objects. For example, walls may be modeled using a generic wall object library in a low-level BIM object throughout the entire model. However, during the detailing stage, external walls that require insulation should be distinguished from internal walls that generally do not require insulation. Such functional requirements based on semantic relationships between objects can be derived using a rule-based approach (Suh 2023), a graph-neural-network-based approach (Park 2023), or other machine-learning approaches.

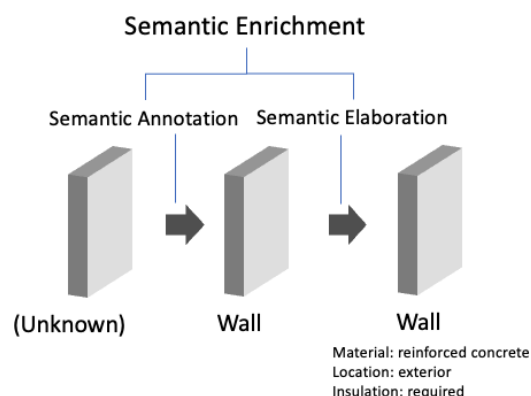


Figure 2. Semantic enrichment, annotation, and elaboration

We categorize semantic enrichment into two types: semantic annotation and semantic elaboration. We use the term, semantic elaboration—a process of adding property information that represents functional requirements—to distinguish semantic annotation—a process of labeling (categorizing) an unknown 2D or 3D shape into a semantically meaningful object such as a wall or a slab (Figure 2).

The third approach is the *large-language-based approach*. This approach develops a detailed library by interacting with a large language model, such as the generative pretrained transformer (GPT) (OpenAI 2023), as a conversational AI and a knowledge base, and replaces the developed library with a target BIM object(s). NADIA is such a system that we developed using GPT as a large language model and an Autodesk Revit addon custom-developed at Yonsei University as a generative design module (Figure 3) (Jang and Lee 2023). A proof-of-concept study shows that NADIA could generate wall details with over 90% accuracy in engineering-wise logical details.

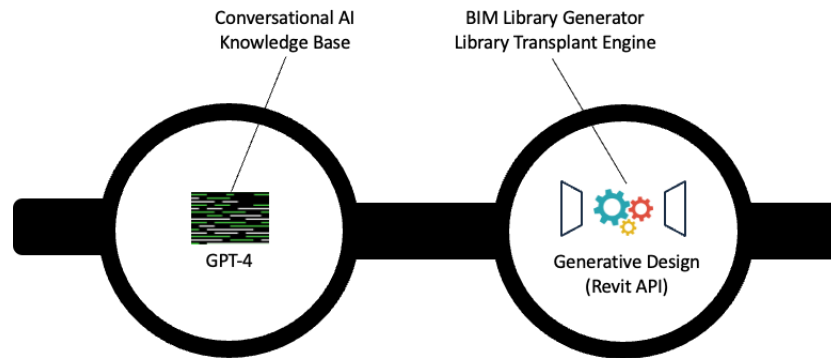


Figure 3. The NADIA framework

CONCLUSION

This paper presents the “building information modeling (BIM) library transplant” framework as an innovative approach to automatically generate detailed BIM models without relying on knowledge-based rules. Three distinct approaches are introduced in this study: attribute matching, semantic elaboration, and the large-language-based approach. The main contributions of this study lie in proposing an innovative framework, overcoming limitations of traditional methods, introducing diverse approaches for BIM library transplant, and demonstrating the potential for more efficient and accurate automated BIM detailing. Although all three approaches demonstrated their feasibility through proof-of-concept experiments, a large room for improvements still remains ahead until they can replace common practices.

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